MelodyMart E-Commerce & Management System

All 6 Members Parts: Features, How to Add Data & Comprehensive Input Validations

Institution: SLIIT (Sri Lanka Institute of Information Technology) | **Module:** SE2030 (Year 2, Semester 1) | **Group ID:** 2026-Y2-S1-MLB-B6G1-08 | **Date:** September 2026

Document Purpose & Architecture Overview:
This official engineering document provides full operational instructions, field specifications, and dual-layer (Frontend HTML5/JS and Backend Servlet/DAO) validation rules for all 6 group members of MelodyMart. All validations are enforced in both the browser client (`pattern` , `min/max` , `step` , `main.js`) and Java backend (`ValidationUtil.java`).

MEMBER 1 OF 6

Module 6.1: User Account & Profile Management

Student Name: Perera D. A. P. W. | **IT Registration Number:** IT25101726 | **Assigned Role:** Customer Authentication, Profile Book & RBAC

How to Add & Use Member 1 Parts:

1. **Customer Registration:** Navigate to `/register.jsp` . Enter Full Name, Email, Sri Lankan Mobile Phone, and Password. Click "*Create MelodyMart Account*" to register.
2. **Address Book Management:** Sign in as a customer, go to `/user?action=profile` . Under "*Delivery Address Book*", click "*+ Add Address*". Fill in Label (Home/Office), Street, City, District, and 5-digit Postal Code. Check "*Primary*" to set default shipping destination.
3. **Admin User CRUD:** Sign in as Administrator (`admin@melodymart.lk` / `Admin@123`). Go to `/admin?tab=users` . Click "*+ Add New System User*". Enter Full Name, Email, Phone, Password, and Role (`CUSTOMER` , `SALES_STAFF` , `INVENTORY_STAFF` , `FINANCE_STAFF` , `ADMIN`). Click "*Register Account in Database*".
4. **Edit / Password Reset:** In the Users table, click "*Edit*" to update contact details, or "*Reset Pwd*" to issue a new secure password.

Member 1 Input Details & Validations Specification

INPUT FIELD	INPUT TYPE	ALLOWED PATTERN / FORMAT	LENGTH / RANGE	VALIDATION RULE & ERROR CODE	EXAMPLE VALID VALUE
`fullName`	`Text`	Letters, spaces, dots, hyphens, apostrophes (`^[A-Za-z\s.'-]+$`)	Min 2, Max 100	Must contain letters and spaces only. `ERR_USER_INVALID_NAME`	`Kasun Chamara Perera`
`email`	`Email`	Standard RFC 5322 format (`^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$`)	Max 100 chars	Unique valid email required. `ERR_USER_INVALID_EMAIL` / `ERR_USER_DUPLICATE_EMAIL`	`kasun.perera@gmail.com`
`phone`	`Tel`	Sri Lankan Mobile (`^(?:0|\+94)?7[0-9]{8}$`)	9 to 12 chars	Restricted to Sri Lankan 07XXXXXXXX or +947XXXXXXXX. `ERR_USER_INVALID_PHONE`	`0771234567` or `+94771234567`
`password`	`Password`	Letters & Numbers (`^(?=.*[A-Za-z])(?=.*\d).{8,}$`)	Min 8, Max 50	Min 8 chars with at least 1 letter and 1 digit. SHA-256 hashed. `ERR_USER_PASSWORD_STRENGTH`	`Melody@2026`
`postalCode`	`Numeric`	5 numeric digits (`^[0-9]{5}$`)	Exactly 5 digits	Must match Sri Lankan 5-digit postal code. `ERR_USER_POSTAL_CODE`	`10115` (Battaramulla)
`street`	`Text`	Alphanumeric address format	Min 3, Max 255	Street address required. `ERR_USER_ADDRESS_REQUIRED`	`142/B, Galle Road`
`city`	`Text`	Letters and spaces (`^[A-Za-z\s.'-]+$`)	Min 2, Max 50	City name required. `ERR_USER_ADDRESS_REQUIRED`	`Colombo`

MEMBER 2 OF 6

Module 6.2: Product Catalog & Advanced Search/Filtering

Student Name: Senarathna U. G. S. P. | **IT Registration Number:** IT25102561 | **Assigned Role:** Catalog Browsing, Search/Filters & Instrument CRUD

How to Add & Use Member 2 Parts:

1. **Add Instrument:** Sign in as Admin or Sales Staff, navigate to `/admin?tab=products`. Click "*+ Add New Musical Instrument*". Enter Title, Category, Brand, Price, Discount, Stock, Description, Technical Specs, Warranty Period (**Years - Numbers Only** e.g. 1, 2, 3), and Image URL. Click "*Register Instrument in Database*".
2. **Edit Instrument:** In the active instruments table, click "*Edit*" to open the pre-populated Edit Modal. Adjust values and click "*Save Changes in Database*".
3. **Delist & Restore:** Click "*Delist*" to archive a discontinued item. In the Archived table, click "*Restore*" to reactivate it.
4. **Catalog Search & Filter:** Open `/catalog.jsp`. Use the search bar for keyword matching (e.g. "*Yamaha*"), filter by category dropdown, brand checkboxes, and minimum/maximum price sliders.

Member 2 Input Details & Validations Specification

INPUT FIELD	INPUT TYPE	ALLOWED PATTERN / FORMAT	LENGTH / RANGE	VALIDATION RULE & ERROR CODE	EXAMPLE VALID VALUE
`name`	`Text`	Descriptive title (`^[A-Za-z0-9\s.'&()-]+$`)	Min 3, Max 150	Instrument name cannot be blank. `ERR_PROD_INVALID_NAME`	`Yamaha FG800 Acoustic Guitar`
`category`	`Select`	String, Keyboards, Drums, Wind, Traditional	Predefined list	Must select valid musical category.	`String Instruments`
`brand`	`Text`	Manufacturer brand name	Min 2, Max 100	Brand name must be at least 2 chars.	`Yamaha`, `Roland`
`price`	`Number`	Decimal currency (`step="0.01"`)	Min LKR 1.00	Price must be positive. `ERR_PROD_INVALID_PRICE`	`78500.00`
`discountPercent`	`Number`	Decimal percentage (`step="0.1"`)	0.0% to 100.0%	Discount must be between 0 and 100%. `ERR_PROD_INVALID_DISCOUNT`	`10.0`
`stockQuantity`	`Integer`	Whole number (`step="1"`)	Min 0 units	Stock units cannot be negative. `ERR_PROD_INVALID_STOCK`	`15`
`description`	`Textarea`	Rich acoustic/material narrative	Min 5, Max 2000	Description must be at least 5 chars. `ERR_PROD_INVALID_NAME`	`Solid Sitka spruce top with scalloped bracing.`
`warrantyPolicy`	`Numeric`	Whole number of years (**Numbers Only**: `^[0-9]+$`)	0 to 25 Years	Warranty period in years (e.g. 1, 2, 3). Characters are strictly forbidden. `ERR_PROD_INVALID_WARRANTY`	`2` (saved as "*2 Years Warranty*")
`imageUrl`	`URL`	Valid HTTP/HTTPS URL (`^https?://.*`)	Max 500 chars	Valid image URL required.	`https://images.unsplash.com/...`

MEMBER 3 OF 6

Module 6.3: Shopping Cart, Checkout & Order Processing

Student Name: Pavishanth K. | **IT Registration Number:** IT25103636 | **Assigned Role:** Real-Time Cart, Voucher Discount & Order Lifecycle

How to Add & Use Member 3 Parts:

1. **Add Items to Cart:** Browse catalog and click "*Add to Cart*". In `/cart.jsp`, update quantities or click "*Remove*".
2. **Apply Voucher Code:** Enter `MELODY10` (10% off) or `WELCOME5` (5% off) and click "*Apply*". Code is auto-capitalized.
3. **Checkout Process:** Click "*Proceed to Checkout*" (`/checkout.jsp`). Fill in recipient name, phone, delivery address, and select District (auto-calculates Sri Lankan delivery fee: Western LKR 500, Outstation LKR 750-950).
4. **Admin Order Management:** In `/admin?tab=orders`, view live customer orders. Click "*Edit Order*" to update recipient name, phone, address, courier partner, payment status, and fulfillment status.

Member 3 Input Details & Validations Specification

INPUT FIELD	INPUT TYPE	ALLOWED PATTERN / FORMAT	LENGTH / RANGE	VALIDATION RULE & ERROR CODE	EXAMPLE VALID VALUE
`quantity`	`Integer`	Whole positive units (`step="1"`)	Min 1, Max stock	Cannot exceed available warehouse stock. `ERR_CART_INVALID_QTY` / `ERR_CART_INSUFFICIENT_STOCK`	`2`
`voucherCode`	`Text`	Alphanumeric, dash, underscore (`^[A-Z0-9_-]+$`)	3 to 20 chars	Voucher must be 3-20 chars, auto-uppercased. `ERR_CART_INVALID_VOUCHER`	`MELODY10`
`customerName`	`Text`	Letters and spaces (`^[A-Za-z\s.'-]+$`)	Min 2, Max 100	Recipient full name required. `ERR_USER_INVALID_NAME`	`Ruwan Jayasuriya`
`shippingAddress`	`Text`	Complete delivery address	Min 5, Max 500	Full address required. `ERR_USER_ADDRESS_REQUIRED`	`No. 45/A, Peradeniya Road, Kandy`

MEMBER 4 OF 6

Module 6.4: Payment Gateway Simulator & E-Invoice System

Student Name: Ekanayaka E. A. T. T. | **IT Registration Number:** IT25101556 | **Assigned Role:** Multi-Channel Payments, Refund/Void & Itemized E-Invoices

How to Add & Use Member 4 Parts:

1. **Credit/Debit Card Checkout:** On checkout, select "*Credit / Debit Card*". Enter 16-digit card number (auto-spaces every 4 digits), Expiry `MM/YY` (auto-inserts slash), and 3-4 digit CVV.
2. **Bank Transfer:** Select "*Bank Transfer*". Enter bank deposit slip receipt reference (e.g. `slip_boc_001.jpg`).
3. **Admin Payment Review:** In `/admin?tab=payments`, click "*Review / Verify Slip*" to inspect bank transfer slips, update status to `PAID` or `FAILED`, and add audit notes.
4. **Process Refund:** Click "*Refund*" on any settled payment. Enter refund amount (LKR), refund reason (min 3 chars), and bank reference code.
5. **Void Payment:** Click "*Void*" on fraudulent or timed-out transactions. Enter reason for cancellation.
6. **Itemized E-Invoice:** Click "*Invoice*" on any transaction row to view and print the auto-generated tax invoice (`/invoice?no=INV-2026-XXXX`).

Member 4 Input Details & Validations Specification

INPUT FIELD	INPUT TYPE	ALLOWED PATTERN / FORMAT	LENGTH / RANGE	VALIDATION RULE & ERROR CODE	EXAMPLE VALID VALUE
`cardNumber`	`Numeric`	16 numeric digits (`[0-9\s]{16,19}`)	Exactly 16 digits	Auto-spaced as XXXX XXXX XXXX XXXX. `ERR_PAY_INVALID_CARD`	`4532 0150 8821 9934`
`cardExpiry`	`Text`	MM/YY format (`(0[1-9]|1[0-2])\/[0-9]{2}`)	5 characters	Auto-slash inserted. Valid month 01-12. `ERR_PAY_INVALID_EXPIRY`	`12/28`
`cardCvv`	`Numeric`	Numeric digits (`^[0-9]{3,4}$`)	3 to 4 digits	Restricted to digits only. `ERR_PAY_INVALID_CVV`	`789`
`bankSlipUrl`	`Text`	Filename with extension (`^[A-Za-z0-9_.-]+$`)	Max 255 chars	Required for bank transfers. `ERR_PAY_UNVERIFIED_SLIP`	`slip_deposit_receipt.jpg`
`refundAmount`	`Number`	Decimal currency (`step="0.01"`)	Min 0.01, Max order total	Refund must be greater than zero. `ERR_PAY_INVALID_AMOUNT`	`5000.00`
`refundReason`	`Textarea`	Detailed reason statement	Min 3, Max 500	Mandatory reason required. `ERR_PAY_REFUND_REASON`	`Customer returned defective bow under warranty.`
`voidReason`	`Text`	Audit cancellation explanation	Min 3, Max 500	Reason must be at least 3 chars. `ERR_PAY_INVALID_VOID`	`Unverified deposit slip / Fraudulent attempt`

MEMBER 5 OF 6

Module 6.5: Real-Time Inventory & Supplier Management

Student Name: Premasiri R. M. R. R. | **IT Registration Number:** IT25103441 | **Assigned Role:** Supplier Directory, Purchase Orders & Stock Auditing

How to Add & Use Member 5 Parts:

1. **Register Supplier:** In `/admin?tab=inventory`, click "+ *Register New Supplier*". Fill in Company Name, Contact Person, Phone, Email, Address, and Categories. Click "*Register Supplier in Database*".
2. **Issue Purchase Order (PO):** Click "+ *Create Purchase Order (PO)*". Select supplier, enter Items Summary, Total Units, Estimated Amount (LKR), and Expected Delivery Date. Click "*Issue Purchase Order*".
3. **Receive Batch & Warranty:** In PO table, click "*Receive Batch*". Select product, verify units, and enter Warranty Batch Tracking Code (e.g. `WB-2026-YMH-8812`).
4. **Stock Delta Adjustment:** In warehouse table, click "*Adjust Stock*". Select type (Add / Restock or Deduct / Damaged), enter Units, and provide mandatory Audit Log Reason.

Member 5 Input Details & Validations Specification

INPUT FIELD	INPUT TYPE	ALLOWED PATTERN / FORMAT	LENGTH / RANGE	VALIDATION RULE & ERROR CODE	EXAMPLE VALID VALUE
`companyName`	`Text`	Company title (`^[A-Za-z0-9\s.,&'()-]+$`)	Min 3, Max 150	Valid supplier name required. `ERR_SUP_INVALID_NAME`	`Yamaha Music Lanka (Pvt) Ltd`
`contactPerson`	`Text`	Letters and spaces (`^[A-Za-z\s.'-]+$`)	Min 2, Max 100	Contact person name (min 2 chars).	`Nimal Perera`
`phone`	`Tel`	Landline or mobile (`^(?:0|\+94)?[0-9\s-]{9,15}$`)	9 to 15 chars	Valid phone number required.	`+94 11 258 7410`
`po quantity`	`Integer`	Positive integer (`step="1"`)	Min 1 unit	PO quantity must be positive. `ERR_PO_INVALID_AMOUNTS`	`25`
`totalAmount`	`Number`	Decimal currency (`step="0.01"`)	Min LKR 1.00	Total amount must be positive. `ERR_PO_INVALID_AMOUNTS`	`1500000.00`
`warrantyBatch`	`Text`	Alphanumeric batch code (`^[A-Za-z0-9_-]+$`)	Min 3, Max 50	Warranty batch code required. `ERR_PO_INVALID_AMOUNTS`	`WB-2026-YMH-8812`
`delta` (Stock)	`Integer`	Whole number delta	Non-zero integer	Resulting stock cannot drop below zero. `ERR_PROD_INVALID_STOCK`	`+10` or `-1`

MEMBER 6 OF 6

Module 6.6: Order Fulfillment, Delivery Tracking & Returns

Student Name: Nellampitiya T. T. | **IT Registration Number:** IT25100652 | **Assigned Role:** Courier Dispatch, Real-Time Tracking & Returns Lifecycle

How to Add & Use Member 6 Parts:

1. **Create Courier Dispatch Job:** In `/admin?tab=fulfillments`, click "+ *Create Courier Dispatch Job*". Select Placed Order from dropdown (auto-fills recipient name, address, and COD amount). Select Courier Partner (`Koombiyo Delivery`, `Domex Courier`, `Pronto Lanka`, `Certis Lanka`). Click "*Dispatch Musical Instruments*". System auto-generates Tracking Number (e.g. `KB-2026-9041`).
2. **Track Shipment:** Open `/order_tracking.jsp`. Enter Tracking Number (e.g. `KB-2026-9041`). Click "*Track Shipment*" to view real-time timeline milestones (`PROCESSING`, `PICKED_UP`, `IN_TRANSIT`, `DELIVERED`).
3. **Submit Return Request:** Open `/returns.jsp`. Enter Order ID, Customer Name, Email, select Reason from dropdown (e.g. "*Defective/Broken Instrument*", "*Damaged in Transit*"), and describe issue in Item Details (min 5 chars). Click "*Submit Return Request*" to generate Ticket ID (e.g. `RET-2026-051`).
4. **Admin Returns Resolution:** In `/admin?tab=fulfillments`, view return tickets table. Update resolution stage (`REQUESTED`, `UNDER_INSPECTION`, `APPROVED`, `REFUNDED`, `REPLACED`) or dismiss ticket.

Member 6 Input Details & Validations Specification

INPUT FIELD	INPUT TYPE	ALLOWED PATTERN / FORMAT	LENGTH / RANGE	VALIDATION RULE & ERROR CODE	EXAMPLE VALID VALUE
`orderId`	`Text / Select`	Alphanumeric order identifier (`^[A-Za-z0-9-]+$`)	Min 3, Max 30	Order ID required for courier dispatch. `ERR_DISP_ORDER_REQUIRED`	`ORD-2026-1001`
`trackingNumber`	`Text`	Courier tracking format (`^[A-Za-z0-9-]+$`)	Min 5, Max 30	Alphanumeric and dashes only. `ERR_DISP_TRACKING_INVALID`	`KB-2026-9041`
`customerName`	`Text`	Letters and spaces (`^[A-Za-z\s.'-]+$`)	Min 2, Max 100	Full recipient name required. `ERR_USER_INVALID_NAME`	`Amila Silva`
`shippingAddress`	`Textarea`	Physical delivery address	Min 5, Max 500	Complete delivery address required. `ERR_USER_ADDRESS_REQUIRED`	`45/A, Peradeniya Road, Kandy`
`codAmount`	`Number`	Decimal currency (`step="0.01"`)	Min LKR 0.00	COD amount cannot be negative. `ERR_DISP_INVALID_COD`	`0.00` or `14500.00`
`reason` (Return)	`Select`	Predefined defect/warranty options	Dropdown option	Must select valid return reason. `ERR_RET_REASON_REQUIRED`	`Defective/Broken Instrument`
`itemDetails` (Return)	`Textarea`	Defect description narrative	Min 5, Max 500	Return details must be at least 5 chars. `ERR_RET_DETAILS_REQUIRED`	`Guitar bridge cracked during transit.`

Summary Matrix: 6 Members Input & Validation Mapping

MEMBER	IT NUMBER & NAME	MODULE TITLE	KEY INPUT FIELDS	VALIDATION METHODS ENFORCED
Member 1	IT25101726 Perera D. A. P. W.	User Account & Profile Management	`fullName`, `email`, `phone`, `password`, `postalCode`, `street`, `city`	`validateUserName`, `validateEmail`, `validateSriLankanPhone`, `validatePasswordStrength`, `validatePostalCode`

MEMBER	IT NUMBER & NAME	MODULE TITLE	KEY INPUT FIELDS	VALIDATION METHODS ENFORCED
Member 2	IT25102561 Senarathna U. G. S. P.	Product Catalog & Search/Filtering	`name`, `brand`, `price`, `discountPercent`, `stockQuantity`, `description`, `imageUrl`	`validateProductName`, `validateBrand`, `validateProductPrice`, `validateProductDiscount`, `validateProductStock`
Member 3	IT25103636 Pavishanth K.	Shopping Cart & Order Processing	`quantity`, `voucherCode`, `customerName`, `customerPhone`, `shippingAddress`	`validateCartQuantity`, `validateVoucherCode`, `validateUserName`, `validateSriLankanPhone`, `validateAddress`
Member 4	IT25101556 Ekanayaka E. A. T. T.	Payment Gateway & Invoices	`cardNumber`, `cardExpiry`, `cardCvv`, `bankSlipUrl`, `refundAmount`, `refundReason`, `voidReason`	`validateCardDetails`, `validatePayment`, `validateRefundReason`, Refund Amount > 0, Void Reason ≥ 3
Member 5	IT25103441 Premasiri R. M. R. R.	Real-Time Inventory & Suppliers	`companyName`, `contactPerson`, `phone`, `email`, `po quantity`, `totalAmount`, `warrantyBatch`, `delta`	`validateSupplier`, `validatePurchaseOrder`, `validateStockDelta`, Warranty Batch Regex
Member 6	IT25100652 Nellampitiya T. T.	Fulfillment, Tracking & Returns	`orderId`, `trackingNumber`, `customerName`, `shippingAddress`, `codAmount`, `reason`, `itemDetails`	`validateDispatch`, `validateTrackingNumber`, `validateUserName`, `validateAddress`, `validateReturnRequest`